# Accessible and scalable pipelines for fast and easy (foodborne) pathogen detection and tracking

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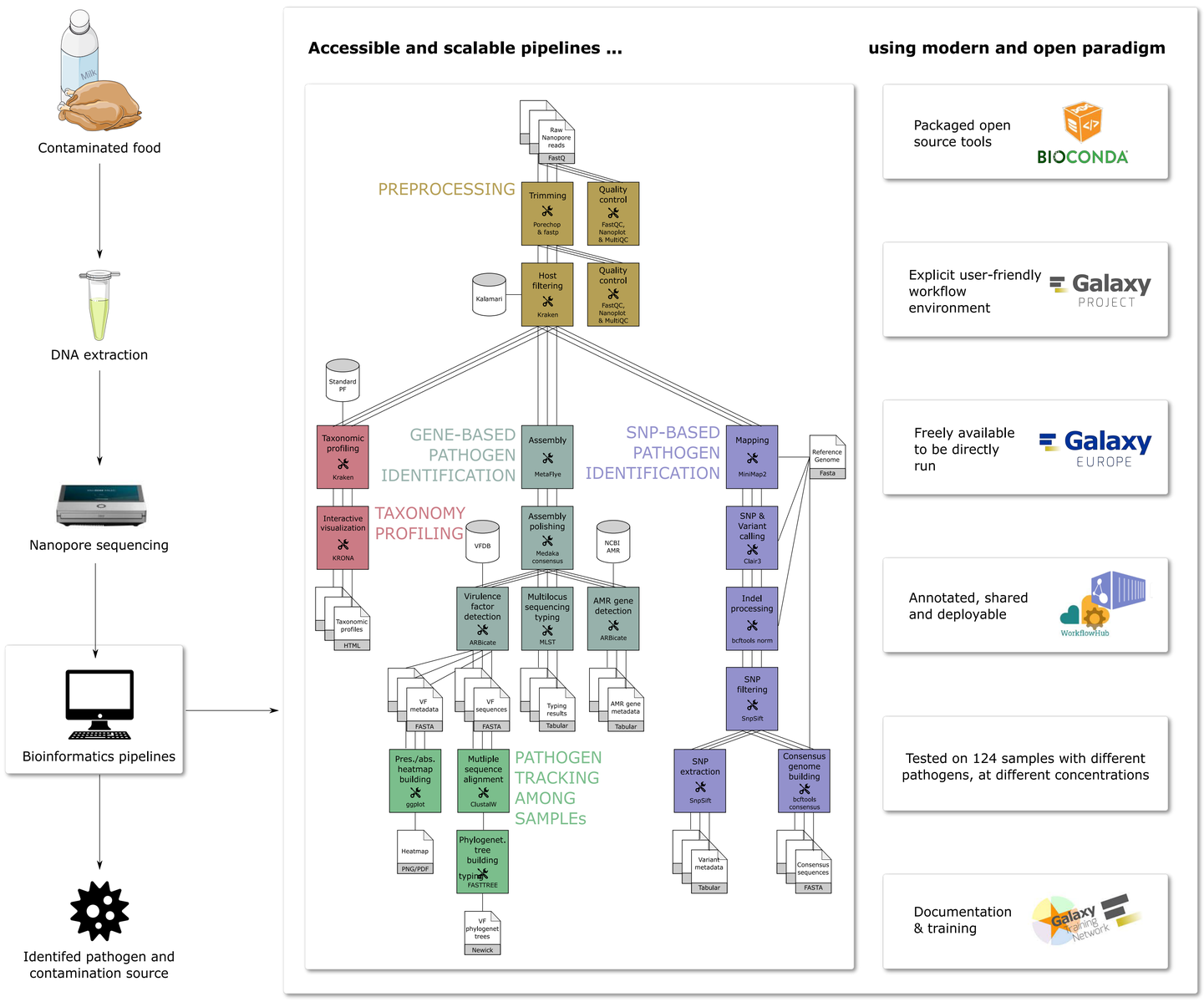
## Context

**Food contamination by pathogens** affects around **600 million people a year** and impacts socioeconomic development at different levels.
During the investigation of a foodborne outbreak, a microbiological analysis of the potentially responsible food vehicle is performed in order to detect the responsible pathogens and identify the contamination source.
**Traditional methods** require the isolation of the targeted pathogen, which is **time-consuming**, not always straightforward, nor successful.
The **metagenomics** approach could solve this issue, by giving an overview of the genomic composition in the sample, including the food source itself, the microbial community, and any possible pathogens. It is **not based on prior DNA isolation**, nor limited to specific genes and it is **more accurate**.
Metagenomics combined with **Oxford Nanopore sequencing** makes the identification of pathogens **quicker**, **easier**, **more accessible**, and **more practical**.
But processing such data stay complex because of the **lack of accessible, easy-to-use, and openly available pipelines**.

## Accessible and scalable pipelines

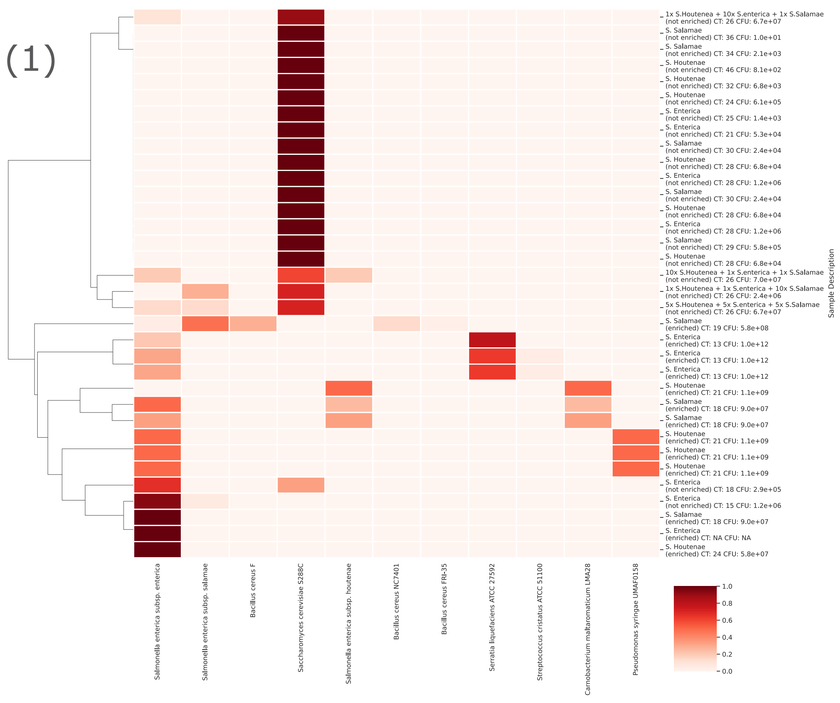
To solve this issue, we have implemented a series of **FAIR Galaxy**-based **workflows**. These workflows integrate **state-of-the-art tools**, **visualization**, and **reports** to **detect** and **track** pathogens from any - not only food - metagenomics Oxford Nanopore sample.




## Results

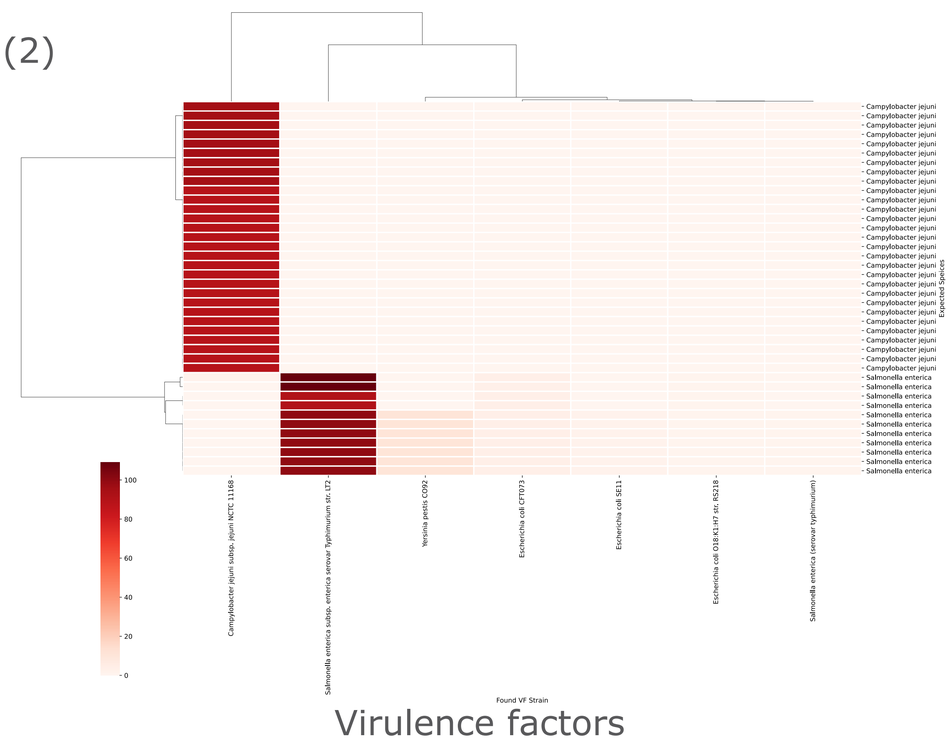
The workflows were **successfully tested** on (1) **spiked** food with different *Salmonella enterica* strains at different concentrations, and
(2) samples collected from human & chicken stools, and meat in Palestine containing *Salmonella enterica* or *Campylobacter jejuni*.

(1)



Subspecies abundances

(2)



Virulence factors

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